An Introduction to Errors and Error Analysis

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Table of contents

Introduction 3
- Resource Credits . . . 4
- Error reporting . . . 4
- Resource history . . . 5

1 How many significant figures are there? 6
- 1.1 Before we begin… Precision and Accuracy. . . . 6
- 1.2 Significant Figures . . . 7
- 1.3 How many significant figures are there? . . . 7
- 1.4 Exercise on counting signficant figures . . . 8
- 1.5 Reading values . . . 8
- 1.6 Round to even . . . 10
- 1.7 Self assessment exercises . . . 11
  - 1.7.1 Self check on our learning so far. . . . 11
  - 1.7.2 Self assessment exercise: Write the following values to the listed sf. . . . 11

2 The use of the correct number of significant figures in calculations. 12
- 2.1 Significant figures in addition and subtraction . . . 12
  - 2.1.1 Exercises with addition and subtraction examples. . . . 13
- 2.2 Significant figures in multiplication and division . . . 15
  - 2.2.1 Exercises with multiplication and division examples. . . . 17
- 2.3 Significant figures in logarithms . . . 18
  - 2.3.1 Exercises with logarythm examples. . . . 19
- 2.4 Significant figures in exponents, powers and roots. . . . 19
  - 2.4.1 Exercises with exponents, powers and roots. . . . 21
- 2.5 Significant figures in mixed calculations . . . 22
  - 2.5.1 Exercises with mixed calculations. . . . 23

3 Averages and standard deviation. 26
- 3.1 Averages and the mean . . . 27
  - 3.1.1 The mean . . . 27
  - 3.1.2 The mid-range . . . 27
  - 3.1.3 Exercises calculating the mean and mid-range. . . . 28
- 3.2 The standard deviation, σ or s. . . . 29
  - 3.2.1 Exercises calculating the mean and mid-range. . . . 32

Introduction

This resource is created to scaffold your learning of errors and uncertainty in data analysis. Whilst these skills are taught here away from the lab, it is usually in the physical and analytical laboratory setting where you will use these skills.

The resource is written in Markdown and has embedded questions which provide feedback. It is strongly encouraged that you start at the beginning and work through on your first use to check your learning. However you should be able to refer to sections as required when you are more familiar with the content.

The resource contains a number of 'highlight blocks'.

> Green blocks contain our 'rules' for handling significant figures and errors.
>
> The rules may appear as text or equations.

> Orange blocks contain worked example questions.
>
> The question followed by commented worked solutions should help you clarify concepts.

> Blue blocks contain questions for you to try. Press the arrow.
>
> Blue blocks will always have the answer 'hidden' in the drop down. Please attempt questions before accessing the drop down, writing down your answer.
> Why? because writing it down make you commit, and if you get it wrong you will think more about why you got it wrong.

> Red blocks contain warnings.
>
> These may be observations about common student errors, or misconceptions.

Chapter Chapter 1

- Section 1.1 Understanding precision and accuracy.
- Section 1.2 Understanding significant figures and how to count them.
- Section 1.3 How many significant figures are there?

- Section 1.5 Making measurements and reading scales.
- Section 1.6 Rounding to even.

Chapter Chapter 2

- Section 2.1 Using significant figures in addition and subtraction.
- Section 2.2 Using significant figures in multiplication and division.
- Section 2.3 Using significant figures with logs.
- Section 2.4 Using significant figures with powers and roots.
- Section 2.5 Using significant figures in mixed calculations.

Chapter 3: Averages and Standard Deviation

- Section 3.1 Calculating averages to appropriate significant figures.
- Section 3.2 Understanding data distribution and standard deviation.

Chapter 4: The Standard Error

- The standard error, a measure of our confidence in the sample mean.
- Use of significant figures in reporting the standard error and the mean.

Chapter 5: The Standard Error of Gradients and Intercepts

- Calculating the standard error of gradients and intercepts in linear regression.
- Examples in Excel.
- Examples in Python.

Resource Credits

This resource was created in conjunction with MAST at the University of Bath. Thanks go to Dr Tamsin Smith and in particular Ed Southwood for considerable assistance in developing the code for the regenerative questions.

This workbook works in conjunction with Chapters 2 and 3 of "An Introduction to Contextual Maths in Chemistry" by Fiona Dickinson and Andrew McKinley, which is available in paper and digital form from the University of Bath library.

Error reporting

Whilst care is taken in production of this manuscript, errors may be present. If you find an error, please report using the form below. Thank you for your help in improving this resource.

Resource history

- 30th July 2026 - Chapter 2 SAQ included.
- 17th July 2026 - First embedded question code included, chapter 2 start.
- 16th June 2026 - First commit of index and chapter 1.

1 How many significant figures are there?

1.1 Before we begin… Precision and Accuracy.

In talking about significant figures, standard erro and propagation of error, we need to be clear about what we mean by precision and accuracy.

Data has a spread, in taking physical measurements this is not something we can avoid. How well we can record a value is frequently limited by the equipment we are using. We will talk more about this spread of data in later chapters of this resource.

Tip 1

- Accuracy is how close a measurement is to the true value.
- Precision is how close a set of measurements are to each other.

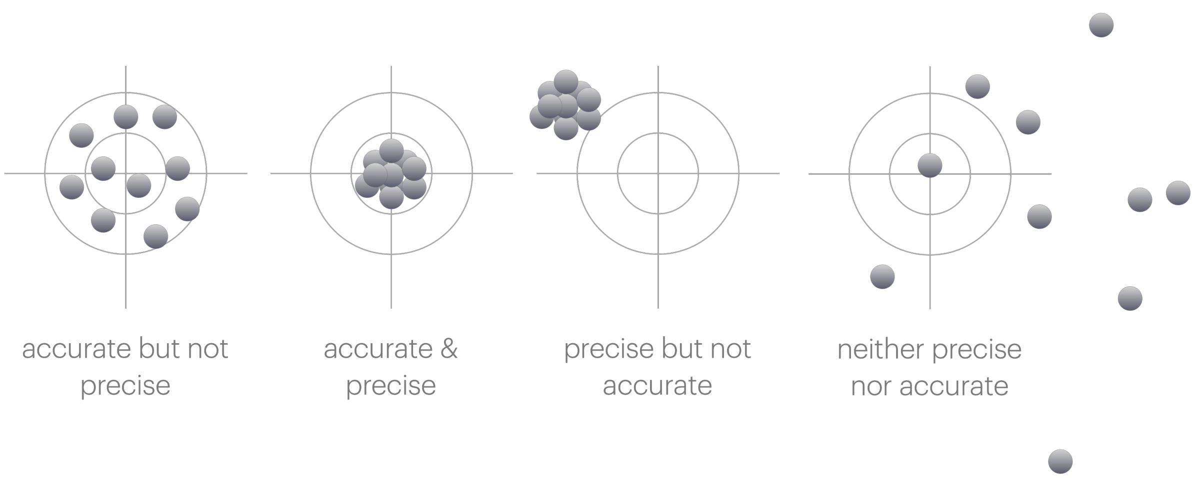


Figure 1.1: The difference between accuracy and precision.

When we are talking about significant figures, we are talking about precision. The more significant figures we have, the more precise our measurement is.

1.2 Significant Figures

Often at school you haven't really thought about significant figures, but in science they are very important. The number of significant figures in a measurement is a measure of how precise that measurement is. The more significant figures, the more precise the measurement.

Therefore it is important that we don't ignore numbers which are significant, or report values to an undue number of significant figures.

Frequently the only way you have thought about significant figures in school is to report things like (2dp) after a value - usually because you have trunkated the value and you want to be clear you are reporting a trunkated value. However this is not appropriate for post school life. Nobody has ever written 2dp or 3sf after a value in a research paper.

> ❗ Don't do this!
>
> Never write 2dp or 3sf after a value. It is not appropriate in science and will make you look like you don't know what you are doing.
> Instead just write your answer to the correct number of significant figures.

Therefore we have to have have a way of understanding the significant figures we have, how to report them in an appropriate way, and then we will see in Chapter 2 how to use them in calculations.

###Why we care about significant figures Significant figures are a measure of our confidence in a value.

When we have many significant figures, we are more confident in the value. When we have few significant figures, we are less confident in the value.

Usually if we have any uncertaintiy in a value it is in the last significant figure we report.

1.3 How many significant figures are there?

Like many things in science, there are rules for how to determine the number of significant figures in a value. The rules are as follows:

> Tip 2
>
> - Any non-zero digit is significant
> - Any 0 following a non-zero digit is significant
> - Any 0 to the left of the first non-zero digit is not significant

By saying that any 0 after another digit we avoid ambiguity. This may be different to what you have been taught in school.

Below are examples of numbers and how many significant figures they have.

How many significant figures are there?

123	3 sf
123.4	4 sf
0.123	3 sf
123400	6 sf
123.4×10^3	4 sf

We should use standard form (or unit prefixes where appropriate) to represent values to an appropriate precision.

! Don't do this!

Freqently in exams I will see answers such as:
$\Delta H = 123854.3628$ J mol^{-1}
≈ 124000 J mol^{-1} (3sf)
This latter value is actually still written to 6 significant figures.
We do not need the $\approx$ sign as if are answer should be to three sig figs then the answer is correct to the correct number of significant figures.
We should write the answer with a unit prefix (or in standard form) to avoid ambiguity.
$\Delta H = 124$ kJ mol^{-1}

1.4 Exercise on counting signficant figures

1.5 Reading values

When we record values in the laboratory we are limited by the precision of the instrumentation we are using. We ususally make decisions about equipment unconciously, for example you don't go seraching for a 4 decimal place balance to weigh out most reagents in the synthetic lab, but you frequently do this in the physical lab.

The balances are a good place to remind us that signficant zeros matter.

There is a big diference between 0.4 g and 0.4000 g in terms of our precision.

One thing that may be different to school is accepting that you cannot read a scale more accurately than the scale allows.

Frequently in school you are told that if it is half way between the graduations on a burrette then you use an increased level of precision than the instrument allows. This is not appropriate in science.

> **Tip 10**
>
> You cannot read a scale more accurately than the scale allows.

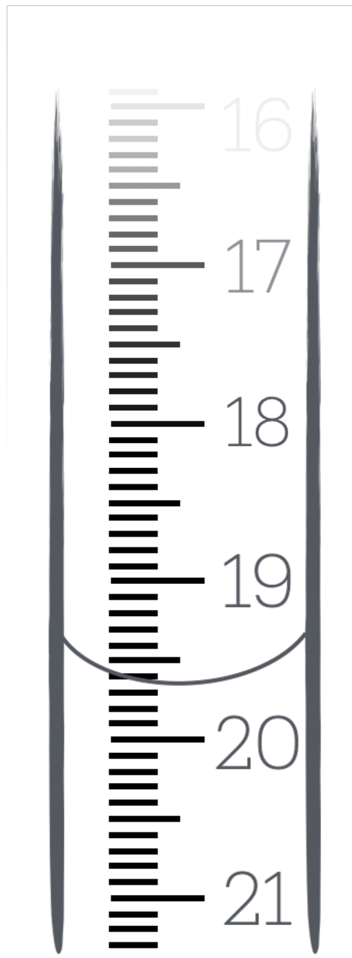


Figure 1.2: The limit of precision on a burette means we can only read as accurately as the scale allows.

In the case of the burette above, we can only read to 1 decimal place. If the meniscus is between 19.6 and 19.7 mL, we **cannot** report the value as 19.65 mL (as you may have been taught in school). This is because we are adding a precision we do not have. How do we know it is 19.65

mL and not 19.66 mL or 19.64 mL? We don't. Therefore we stick with the precision of the instrument.

In this case you either have to choose 'is it closer to 19.6 mL or 19.7 mL' and report the value as either 19.6 mL or 19.7 mL or to avoid systematic bias we *round to even*.

1.6 Round to even

If we think about rounding we need to consider the possibility of systematic bias. If we always round up, then we will always be reporting a value that is higher than the true value. If we always round down, then we will always be reporting a value that is lower than the true value.

Consider the following values which we need to report to 1 decimal place:

Rounding values?

19.10	19.1	no rounding needed
19.11	19.1	round down
19.12	19.1	round down
19.13	19.1	round down
19.14	19.1	round down
19.15	19.?	???
19.16	19.2	round up
19.17	19.2	round up
19.18	19.2	round up
19.19	19.2	round up
19.20	19.2	no rounding needed

If we always round the value ending in a 5 up (as you maybe always have before) then we have more values rounding up than we have rounding ndown and we introduce a slight systematic bias increasing our value over all.

Therefore if a value ends exactly in a 5 and needs to be rounded we instead 'round to even', which should mean that half of the time it will round up and half of the time it rounds down.

Tip 4

If a value ends in a 5 exactly and needs rounding we should round to the nearest even number.

Examples of rounding to even.

If each of the following should be reported to 3 significant figures

47.58	rounds up to	47.6	
8.325	rounds down to	8.32	rounded to even
3.621	rounds down to	3.62	
64.55	rounds up to	64.6	rounded to even
10.050003	rounds up to	10.5	

1.7 Self assessment exercises

1.7.1 Self check on our learning so far.

What is 16453 rounded to 2 significant figures?

16×10^3
Any number we write is considered significant so we can't write 16000 and think it is 2 significant figures. We need to use standard form to avoid ambiguity.

What is 0.090305 rounded to 2 significant figures?

0.090
It is easy to forget significant zeros after our first non-zero digit. In this case we have 2 significant figures, the 9 and the 0 after it.

What is 2.4503 rounded to 2 significant figures?

2.5
In this case round to even doesn't apply as it isn't exactly '5'.

What is 37.4500 rounded to 3 significant figures?

37.4
In this case round to even does apply as it is a 5 we are having to round.

1.7.2 Self assessment exercise: Write the following values to the listed sf.

2 The use of the correct number of significant figures in calculations.

In Chapter 1 we have learnt about significant figures and why they are important in understanding the confidence we have in a value.

Now we need to learn the rules to how many significant figures we should use after doing calculations.

There are different rules for how we handle significant figures depending upon the type of calculation we have done. Freqently we may have more than one step in a calculation, and therefore we need to be concious of the appropriate significant figures in each step.

2.1 Significant figures in addition and subtraction

When adding or subtracting values, the number of decimal places in the result should be the same as the value with the least number of decimal places in your calculation.

We cannot add precision, I can't measure something with a meter rule with 1 cm increments, a 15 cm rule with 1 mm increments and a micrometer with 1 m increments and then add them together and expect to have a result with 1 m precision. The result will be limited by the least precise measurement.

> Tip 10: Significant figures in addition and subtraction.
>
> When adding or subtracting values, the number of decimal places in the result should be the same as the value with the least number of decimal places in your calculation.

> Examples of significant figures in addition and subtraction.
>
> - Convert 20 ºC to Kelvin.
>
> We know our measured temperature to 0 decimal places, therefore our answer should be to 0 decimal places.
> T = 20 ºC + 273.15 K = 293 K
>
> - An ideal gas was heated with 1.75 kJ of energy an allowed to do 250 J of work.

What is the change in internal energy of the gas?

In this case we have different units, so if we convert 250 J to kJ we have 0.250 kJ. The least number of decimal places is 2, therefore our answer should be to 2 decimal places.
$\Delta U = 1.75 \text{ kJ} - 0.250 \text{ kJ} = 1.50 \text{ kJ}$
This unit conversion kJ to J (or vice versa) is common in thermodynamics calculations where ΔH is usually reported in kJ mol^{-1}, but ΔS is usually reported in J K mol^{-1}.
It is easy to forget the significant zero at the end of this number as usually our calculator will not report it, but including it gives us our answer to a higher, more appropriate, precision.

2.1.1 Exercises with addition and subtraction examples.

2.1.1.1 Self check on our learning so far.

What is 15.4 g + 0.16 g to the correct number of significant figures?

15.6 g
The rule for adding and subtracting is that the answer should be reported to the least number of decimal places in the calculation. In this case 15.4 g has 1 decimal place, and 0.16 g has 2 decimal places, so our answer should be reported to 1 decimal place.

What is -154.3 kJ mol^{-1} + 894.2 J mol^{-1} to the correct number of significant figures?

-153.4 kJ mol^{-1}
Here we have different units, so if we convert 894.2 J mol^{-1} to kJ mol^{-1} we have 0.8942 kJ mol^{-1}.
Now we can compare the number of decimal places in each value. -154.3 kJ mol^{-1} has 1 decimal place, and 0.8942 kJ mol^{-1} has 4 decimal places, so keeping the fewest decimal places, our answer should be reported to 1 decimal place.

The following are molar masses of some common elements. Use these values to answer the following questions.

Table 2.1: A selection of atomic masses of common elements.

	R_A / g mol^{-1}
H	1.00794
C	12.011
O	15.9994

	R_A / g mol^{-1}
N	14.0067
Cl	35.45

What is the molar mass of HCl to the correct number of significant figures?

36.46 g mol^{-1}
35.45 g mol^{-1} + 1.00794 g mol^{-1} = $\cancel{36.45794 \text{ g mol}^{-1}}$ = 36.46 g mol^{-1}
Cl has the least number of decimal places (2), so our answer should only be reported to 2 decimal places.

Given we should now be in the habit of reporting our answers to the correct number of significant figures, this is no longer going to be asked in the following exercises, but we should still be concious of the number of significant figures in our answers.

What is the molar mass of C_2H_5OH?

46.064 g mol^{-1}
Carbon's molar mass has 3 decimal places, hydrogen's molar mass has 5 decimal places and oxygen's molar mass has 4 decimal places.
In addition questions our answer should be reported to the fewest decimal places, which in this case is 3.

What is the molar mass of $C_6H_{12}O_6$?

180.158 g mol^{-1}
Once you get used to the idea of handling signficant figures you will be able to perform your calculations, then correct your values to the correct number of significant figures when reporting.
Here Carbon has 3 decimal places, so the answer should be reported to 3 decimal places, even if the value coming from a calculator would be 180.15768.

What is the molar mass of $C_{12}H_8N_2$?

180.209 g mol^{-1}
Again carbon limits the number of decimal places to give in our answer.

What is the molar mass of $C_{12}H_{22}O_{11}$?

342.300 g mol^{-1}
It is important to remember to include significant zeros if they are there. These are easy to remember because there were non zero numbers after them, but remember to write them down.

2.1.1.2 Self assessment question on significant figures in addition and subtraction.

Please remember you can refresh this question.

2.2 Significant figures in multiplication and division

When multiplying or dividing values, the number of significant figures in the result should be the same as the value with the least number of significant figures in your calculation.

First we need to quickly count the number of significant figures in each value. If we have a unit conversion (such as mm to cm) then this is 'exact' and does not change the number of significant figures in our value. Some unit conversions have limited significant figures, others are precise. In either case it is always a good habit to use more significant figures in constants than in our other values.

Tip 10: Significant figures in multiplication and division.

When multiplying or dividing values, the number of significant figures in the result should be the same as the value with the least number of significant figures in your calculation.

Example of significant figures in multiplication and division - the ideal gas law.

- Determine the number of moles of gas in a cylinder with a volume of exactly 2.52 L at a pressure of 50.24 bar and a temperature of 273 K.

Here we need to do a couple of unit conversions to get our values into base SI units.
The volume is already in L, which is a derived unit, but we need to convert it to m^3. 1 L $\equiv$ 1 dm^3 = 1 x 10^{-3} m^3, so we have 2.52 L = 2.52 x 10^{-3} m^3. This is an exact conversion, so the number of significant figures in our value does not change. Recall that the volume is described as "exactly" 2.52 L, so we have 'infinite' significant figures in our volume.
The pressure is in bar, which is not an SI unit, so we need to convert it to Pa. 1 bar = 1 x 10^5 Pa, so we have 50.24 bar = 5.024 x 10^6 Pa. This is also an exact conversion. The pressure has 4 significant figures, so our pressure in bar will also have 4 significant figures.

Using the ideal gas law, we can calculate the number of moles of gas in the cylinder.
$pV = nRT$ so $n = \frac{pV}{RT}$
Putting in our values:

$$n = \frac{5.024 \times 10^6 \text{ Pa} \times 2.52 \times 10^{-3} \text{ m}^3}{8.31446 \text{ J K}^{-1} \text{ mol}^{-1} \times 273 \text{ K}}$$

$n = 5.58$ mol
Temperature only has 3 significant figures, so our answer should be reported to 3 significant figures.

Example of significant figures in multiplication and division - heat capacity.

- Determine the energy put into a system when a 1.000 kg block of copper is heated from 20.16 ºC to 60.52 ºC. The specific heat capacity of copper is 385 J kg^{-1} K^{-1}.

We can calculate the energy put into the system using the equation:
$q = mc\Delta T$
First we need to determine the change in temperature, ΔT.
$\Delta T = T_{\text{final}} - T_{\text{initial}}$
$\Delta T = 60.52C - 20.16C = 40.36C$
We've followed the rules for addition and subtraction, so our answer has 2 decimal places, which is the same as the least number of decimal places in our calculation.
Now we can calculate the energy put into the system, q.
$q = 1.000 \text{ kg} \times 385 \text{ J kg}^{-1} \text{ K}^{-1} \times 40.36 \text{ K}$
The mass has 4 significant figures, the specific heat capacity has 3 significant figures, and the temperature change has 4 significant figures. The least number of significant figures in our calculation is 3, so our answer should be reported to 3 significant figures.
$q = 15.5$ kJ
Note: I've changed my units to kJ to make reporting to 3 significant figures easier. If I had left my answer in J, I would have reported it as 15.5×10^3 J, which is not as clear (and is more to type).

2.2.1 Exercises with multiplication and division examples.

2.2.1.1 Self check on our learning so far.

If each side of a rectangle was measured to be 2.34 cm and 1.2 cm, what is the area?

A = 2.8 cm^2
The area of a rectangle is given by $A = l \times w$, so from our calculator we get a value of 2.808 cm^2. Since one of our sides only has 2 significant figures, using our rule of using the fewest significant figures, our answer should be reported to 2 significant figures.

What is the volume of a cylinder with a diameter of 2.3 cm and a height of 12.4 cm?

V = 52 cm^3
On the surface this is an easy calculation following our rules of significant figures we retain the fewest significant figures in our answer, which is 2.
You might be tempted to use the radius of the circle, and 'half' the diameter, but in doing this we can get into a mess with our significant figures because we can't gain significant figures by dividing by 2.
Instead we can use the diameter directly in our calculation, $V = \frac{\pi d^2 h}{4}$ which simplifies the significant figure handling.
This shows that is is often easiest to use algebraic rearrangements to avoid having to do extra calculations which may introduce complications in our consideration of significant figures.

Determine the rate constant of diffusion, k_d at 278 K, if the viscosity of the medium, η, is 1.518 mPa s, given $k_d = \frac{8RT}{3\eta}$.

$k_d = 4.06 \times 10^6$ m^3 s^{-1} = 4.06 $\times 10^9$ dm^3 s^{-1}
This is a more challenging question as you won't hve come across this equation before, but the principles are the same. This is a multiplication and division question, so we need to consider the number of significant figures in each value, and report our answer to the fewest number of significant figures in our calculation.
The integers, 8 and 3 are exact values, so can be considered to have infinite significant figures.
When values are large some poeple want to write 4 060 000 (3sf) but given our rule that if you write a digit it is significant this answer is actually 7 significant figures.

2.2.1.2 Self assessment question on significant figures in multiplication and division.

Please remember you can refresh this question.

2.3 Significant figures in logarithms

Significant figures in logs and lns are a little different to the rules for addition, subtraction, multiplication and division because we have to think about the 'value' when accounting for how many significant figures we have in our answer.

Tip 10: Significant figures in logs and lns.

When taking logs we usually gain 1 significant figure, however if the 'logged value' is between 0 and the base then we do not gain a significant figure.

Practically what does this mean? and why the do we sometimes gain a significant figure? The following example using $\log_{10}$ will help to explain this.

Example of significant figures in logs and lns.

Determine $\log(2.234)$, $\log(22.34)$ and $\log(223.4)$ to the correct number of significant figures.
Our first example is $\log(2.234)$. The value of $\log(2.234)$ is 0.3490831688....
The value we are taking the log of has 4 significant figures, so our answer should have 4 significant figures. Therefore we report our answer as 0.3491.
Now, if I take the log of 22.34 I can also write this as $\log(2.234 \times 10^1)$.
Using our 'rules of logs', we can write this as $\log(2.234) + \log(10^1) = 0.3491 + 1 = 1.3491$.
Since our value of $\log(10^1)$ is an exact value, it can be considered to have infinite significant figures, and we can add these values according to the rules in Section 2.1.
Finally, if I take the log of 223.4 I can also write this as $\log(2.234 \times 10^2)$. The value we are taking the log of has 4 significant figures, so our answer should have 4 significant figures. Therefore we report our answer as 2.3491.
You can see why we gain a significant figure when taking the log of a value greater than 10, but not when taking the log of a value less than 10. The 'value' of the number we are taking the log of is important in determining how many significant figures we have in our answer.
The same principle works if we have very small numbers, for example $\log(0.002234)$ which we can write as $\log(2.234 \times 10^{-3}) = \log(2.234) + \log(10^{-3}) = 0.3491 - 3 = -2.6509$.

2.3.1 Exercises with logarythm examples.

2.3.1.1 Self check on our learning so far.

Calculate log 5.34.

0.728
Because 5.34 is between 1 and the base (in this case base 10) we do not gain a significant figure in our calculation.

Calculate ln 7.156.

1.9680
In this case we do gain a significant figure as our value is greater than the base (in this case base e). A quick way to know if we are bigger than the base is if our calculated logarithm is greater than 1.

Determine the pH of a 5.59 µM solution of a strong acid. You may assume the activity coefficient is 1.

pH = 5.253
pH $= -\log\{H^+\}$
Our concentration may also be written as 5.59×10^{-6}, this value is less than 1, so we will gain a significnant figure.

2.3.1.2 Self assessment question on significant figures with logarythms.

2.4 Significant figures in exponents, powers and roots.

Tip 10: Significant figures with powers and roots.

Squaring, cubing, or rooting a value our answer should have the same number of significant figures as the value we operated on.

This is easiest to think about in terms of squaring or cubing, where we can think about things in terms of multiplying values.

Example of significant figures in powers - volume.

- Determine the volume of a sphere with diameter 4.52 cm.

Again, I'm going to use the equation for volume with diameter. The volume of a sphere is given by: $V = \frac{4}{3}\pi r^3$, this can be written in terms of diameter, d, as: $V = \frac{1}{6}\pi d^3$
π is a number with 'infinite' significant figures.
So in our calculation our volume will carry the same number of signifiant figures as the diameter.
V = 48.4 cm^3

This principle stands with any power, be it integers, fractions (square and cube roots), or negative powers. The number of significant figures in our answer should be the same as the number of significant figures in the value we are operating on.

If you recall, in some circumstances when taking logarythms we sometimes gained a significant figure. When taking exponents (the reciprocal function of logs) then this means that we sometimes need to loose a significant figure.

Tip 10: Significant figures with exponents.

When taking an antilog (so for example 10^x or e^x) we should retain the same number of significant figures if our exponent is between -1 and 1.
If our exponent is less than -1 or greater than 1 then we should loose a significant figure.

Why is the rule between -1 and 1?

This is because we can write negative exponents as positive exponents in a fraction, so $e^{-x} = \frac{1}{e^x}$.

For ease this first example is 'undoing' one of our examples from Section 2.3.

Example of significant figures exponents.

Determine $10^{0.3491.}$ and $10^{1.3491}$ to the correct number of significant figures.
This is the type of question you see if calculating activity from pH.
For our first example $10^{0.3491}$ our exponent is between 0 and 1, so we should retain the same number of significant figures in our answer.
$10^{0.3491} = 2.234$
However, for our second example $10^{1.3491}$ our exponent is greater than 1, so we should loose a significant figure in our answer, so 5 significant figures in our exponent gives us 4 significant figures in our answer.
$10^{1.3491} = 22.34$

2.4.1 Exercises with exponents, powers and roots.

2.4.1.1 Self check on our learning so far.

Calculate $e^{0.728}$.

2.07
Because our power is between 0 and 1, we should retain the same number of significant figures in our answer.

Calculate $e^{-0.38}$.

0.68
Because our power is between -1 and 1, we should retain the same number of significant figures in our answer.
Rembember when we have a negative exponent we can write it as a positive exponent in a fraction, so $e^{-0.38} = \frac{1}{e^{0.38}}$.

Calculate $10^{4.5718}$.

37.31×10^3
This is one of the cases where our exponent is greater than 1, so we should loose a significant figure in our answer, so 5 significant figures in our exponent gives us 4 significant figures in our answer.
Because writing our answer as 37310 would contain 5 significant figures, (even if we think we've rounded it to 4 significant figures) we need to write our answer in scientific notation to show that we have 4 significant figures in our answer.

Calculate $10^{-6.18}$.

6.6×10^{-7}
This is one of the cases where our exponent is less than -1, so we should loose a significant figure in our answer, so 3 significant figures in our exponent gives us 2 significant figures in our answer.

Calculate $10^0.477$.

3.0
We have lost a significant figure, but we need to remember to include the significant zero in our answer.

Now a chemistry example

Determine the activity of H^+ in a solution with pH = 5.25.

$\{H^+\} = 5.6 \times 10^{-6}$
Recall that $\mathrm{pH} = -\log\{H^+\}$, so we can rearrange this to give us $\{H^+\} = 10^{-\mathrm{pH}}$.
We have a negative exponent, and it is less than -1, so we should loose a significant figure in our answer, so 3 significant figures in our exponent gives us 2 significant figures in our answer.
Note: curly brackets are used to denote activity, and square brackets are used to denote concentration.

2.5 Significant figures in mixed calculations

Many calculations we see in chemistry involve a mixture of different mathematical operators, which you have learned to handle in school using the rule of BODMAS/BIDMAS (Brackets, Order/Indices, Division and Multiplication, Addition and Subtraction).

Tip 10: Significant figures in mixed calculations.

When we have a mixed calculation we need to follow the appropriate number of significant figures appropriate in a stepwise (BIDMAS/BODMAS) manner, and then report our final answer to the appropriate number of significant figures.

When we have a mixed calculation we need to follow the appropriate number of significant figures appropriate in a stepwise manner, and then report our final answer to the appropriate number of significant figures.

Practically we do not 'round' our answer at each step, (because of the probability of making an error in our input) and only round our final answer to the appropriate number of significant figures.

Common examples of mixed calculations in chemistry involve temperature, where the conversion between Celsius and Kelvin is frequently used. Thermodynamics also gives us $\Delta G = \Delta H - T\Delta S$, further kinetics calculations frequently involve both multiplication and either logs or powers.

Examples of significant figures in mixed calculations.

Determine the change in Gibbs free energy, ΔG, for a reaction at 25 K, if $\Delta H = -84.3$ kJ mol^{-1} and $\Delta S = 0.164$ kJ K^{-1} mol^{-1}.
$\Delta G = \Delta H - T\Delta S$
The first step we have to do here is to calculate T in K. Although this isn't formally in a bracket, we need to do this first to get our temperature in the correct units. This is an

addition calculation, so we need to consider the number of decimal places in our answer. So $T = 25$ ºC + 273.15 K = 298 K (we should be considering this value to have 3 sig figs). The next step is to calculate $T\Delta S$. This is a multiplication calculation, so we need to consider the number of significant figures in our answer.
So $T\Delta S = 298 \text{ K} \times 0.164 \text{ kJ K}^{-1} \text{ mol}^{-1} = 48.9 \text{ kJ mol}^{-1}$
The final step is to calculate $\Delta G = \Delta H - T\Delta S$. This is a subtraction calculation, so we need to consider the number of decimal places in our answer.
So $\Delta G = -84.3 \text{ kJ mol}^{-1} - (48.9 \text{ kJ mol}^{-1}) = -133.2 \text{ kJ mol}^{-1}$
This is potentially a confusing result if you don't have a good understanding of significant figures, as the final answer has more significant figures than the values we used in our calculation.

Examples of significant figures in mixed calculations.

Determine the rate constant for the first order reaction where the initial concentration of the reactant is 0.100 M, the final concentration of the reactant is 0.127 M and the time taken for this change is 70 minutes.
If it is a first order reaction, then we can use the equation:
$[A] = [A]_0 e^{-kt}$
which we can rearrange to give us:
$k = \frac{1}{t} \ln \frac{[A]_0}{[A]}$
The first step is to calculate the ratio of the initial and final concentrations. This is a division calculation, so we need to consider the number of significant figures in our answer.
$\frac{[A]_0}{[A]} = \frac{0.100 \text{ M}}{0.127 \text{ M}} = 0.787$
The next step is to calculate the natural log of this value. This is a log calculation, so we need to consider the number of significant figures in our answer.
$\ln \frac{[A]_0}{[A]} = \ln(0.787) = -0.2395$
In this case we are taking the log of a value less than 1 so we should gain a significant figure, so our answer has 4 significant figures.
The final step is to calculate the rate constant k. This is a multiplication calculation, so we need to consider the number of significant figures in our answer.
$k = \frac{1}{t} \ln \frac{[A]_0}{[A]} = \frac{1}{70 \text{ min}} \times (-0.240) = -0.0034 \text{ min}^{-1}$
Here our time, t, has the fewest significant figures, so our answer should be reported to just 2 significant figures.

2.5.1 Exercises with mixed calculations.

2.5.1.1 Self check on our learning so far.

In each case the equation you will need to use is given above the question, but it may be in a form that you need to rearrange to get the value you are looking for.

$\Delta G = \Delta H - T\Delta S$.

ℹ Determine the Gibbs free energy change at 50 ºC for the formation of calcite (CaCO—3), given the enthapy of formation is -1207 kJ mol^{-1} and the entropy of formation is +92.9 J K^{-1} mol^{-1}.

$\Delta G = -1237$ kJ mol^{-1}
To do this we need the equation $\Delta G = \Delta H - T\Delta S$.
First we need to convert our temperature to Kelvin, which is an addition calculation, so we need to consider the number of decimal places in our answer.
T = 50 ºC + 273.15 K = 323 K (we should be considering this value to have 3 sig figs).
Next we need to calculate $T\Delta S$. This is a multiplication calculation, so we need to consider the number of significant figures in our answer.
$T\Delta S = 323$ K $\times$ 92.9 J K^{-1} mol^{-1} = 3.00×10^4 J mol^{-1} = 30.0 kJ mol^{-1}
Finally we can calculate $\Delta G = \Delta H - T\Delta S$. This is a subtraction calculation, so we need to consider the number of decimal places in our answer.
$\Delta G = -1207$ kJ mol^{-1} $-$ 30.0 kJ mol^{-1} = -1237 kJ mol^{-1}

$\Delta G = -RT \ln K$

ℹ Determine the equilibrium constant, K, for a reaction with a Gibbs free energy change of -10.45 kJ mol^{-1} at 36.5 ºC.

$K = 57.9
The first step here is to rearrange the equation to give us K as the subject.
$K = e^{-\frac{\Delta G}{RT}}$
Now we need to convert our temperature to Kelvin, which is an addition calculation, so we need to consider the number of decimal places in our answer.
T = 36.5 ºC + 273.15 K = 309.6 K (this value has been rounded to even and has 4 significant figures).
Next we need to calculate the value of $\frac{\Delta G}{RT}$. This is a multiplication/division calculation, so we need to consider the number of significant figures in our answer.
$\frac{\Delta G}{RT} = \frac{-10.45 \text{ kJ mol}^{-1}}{8.31447 \text{ J K}^{-1} \text{ mol}^{-1} \times 309.6 \text{ K}}$
rewriting our units of kJ to J we have:
$\frac{\Delta G}{RT} = \frac{-10.45 \times 10^3 \text{ J mol}^{-1}}{8.31447 \text{ J K}^{-1} \text{ mol}^{-1} \times 309.6 \text{ K}}$
$\frac{\Delta G}{RT} = -4.060$ - our fewest significant figures is 4, so our answer should be reported to 4 significant figures.
now we can calculate $K = e^{-\frac{\Delta G}{RT}}$.
$K = e^{4.060} = 57.9$ - because our exponent is greater than 1, we should loose a significant figure in our answer, so 4 significant figures in our exponent gives us 3 significant figures in our answer.
Finally we can calculate $K = e^{-\frac{\Delta G}{RT}}$. This is an exponent calculation, so we need to

consider the number of significant figures in our answer.

$E = E^0 - \frac{RT}{nF} \ln \frac{\{\text{products}\}}{\{\text{reactants}\}}$

Recall - values in curly brackets are activities.

Determine the cell potential, E for a reaction at exactly 20 ºC, with a standard cell potential, E^0, of 1.012 V, a number of electrons transfered, n, of 2, and activities of the products and reactants of 0.563 and 1.263 respectively.

$V = 1.022$ V

First we need to deal with the 'implied' brackets:

$\frac{\{\text{products}\}}{\{\text{reactants}\}} = \frac{0.563}{1.263} = 0.446$

Three significant figures in our answer because our fewest significant figures in our calculation is 3.

$T = 20C + 273.15K = 293.15K$

The temperature in the question is described as "exactly" 20 ºC, so we can consider this to have infinite significant figures. Therefore our temperature in Kelvin will also have infinite significant figures.

The next highest priority operation is the log:

$\ln \frac{\{\text{products}\}}{\{\text{reactants}\}} = \ln(0.446) = -0.807.$

The value we are taking the log of is between 0 and the base, so we do not gain a significant figure in our answer.

The next highest priority operation is the multiplication/division:

$\frac{RT}{nF} \ln \frac{\{\text{products}\}}{\{\text{reactants}\}} = \frac{8.31447 \text{ J K}^{-1} \text{ mol}^{-1} \times 293.15 \text{ K}}{2 \times 96485 \text{ C mol}^{-1}} \times (-0.807) = -0.0102 \text{ V}$

The final step is to calculate the cell potential, E. This is a subtraction calculation, so we need to consider the number of decimal places in our answer.

$E = E^0 - \frac{RT}{nF} \ln \frac{\{\text{products}\}}{\{\text{reactants}\}} = 1.012 \text{ V} - (-0.0102 \text{ V}) = 1.022 \text{ V}$

2.5.1.2 Self assessment question on mixed calculations.

3 Averages and standard deviation.

You are likely familiar with the concept of averages, and you may have heard of the standard deviation. In this chapter we will look at how to calculate averages and standard deviations, and how to report them to the appropriate number of significant figures.

It is important to remember that all data has a spread, we do not expect every measurement to be exactly the same no matter how careful we are. The spread of the data is a measure of the uncertainty in our measurements, and we can use that to calculate the standard error.

Figure 3.1: The spread of data.

Your data being ‘far’ from the mean isn’t a bad thing, it just means that you have collected data that falls within the expected spread of the data. This distrubtion of data is why, as scientists, we take multiple measurements of the same thing. We want to be sure that our data is representative of the true value.

It is also why we take many measurements of the same thing, and then calculate an average value. The average value is a better representation of the true value than any single measurement.

You will notice from Figure 3.1 that the data points are symmetrically distributed around the mean. This is what we expect to see in Gaussian (sometimes called normal) distribution of data, you will likely have seen this distrubtion in school, or be familiar with the concept of standard deviation. When we calculate a standard deviation (and later a standard error) we are assuming that our data is normally distributed, and that the mean is a good representation of the data.

3.1 Averages and the mean

3.1.1 The mean

There are different types of average, but the one we are interested in, and the one you are probably most familiar with, is the mean. The mean is calculated by adding all the values together and dividing by the number of values.

Tip 11: Calculating the mean, $\bar{x}$.

$\bar{x} = \frac{\sum_{i=1}^{N} x_i}{N}$

The overline above the x indicates it is the mean of the x values, we can use this notation for any variable, for example the mean of ΔT would be written as $\bar{\Delta T}$, or $\Delta_{\bar{r}} H$.

- When calculating the mean using Excel you can use the function '=AVERAGE()' to calculate the mean of a set of values.
- When calculating the mean using Python, you can use the `statistics.mean()` or `numpy.mean()` function, after importing the necessary libraries.

When we calculate the mean we need to remember the rules for significant figures from Chapter 2. The mean is a calculated value, and therefore the number of significant figures in the mean is determined by the number of significant figures in the values we are averaging.

Calculating the mean of a set of data.

Determine the mean of the following set of data: ΔT / K : 10.33, 10.42, 10.41, 10.60, 10.57, 10.33, 10.49
$\bar{\Delta T} = 10.45$ K
$\bar{T} = \frac{\sum_{i=1}^{N} T_i}{N}$
here N = 7, and $\sum_{i=1}^{N} x_i = 73.15$ K
$\bar{\Delta T} = \frac{73.15 \text{ K}}{7} = 10.45$ K

3.1.2 The mid-range

The mid-range is another type of average, which can be calculated very quickly to give you an estimate of the average value.

Tip 12: Calculating the mid-range.

Mid-range $= \frac{\text{Max value}+\text{Min value}}{2}$

Like the mean our confidence in the mid-range increases with the number of data points we have collected. If we collect a large amount of data, the mid-range will be a good estimate of the mean. However if we have only collected a small amount of data, the mid-range will likely not be a good estimate of the mean.

Also like the mean, we cannot gain significant figures during our calculation.

In practice we don't use the mid-range very often, but it is a quick way to get an estimate of the mean value of a set of data in the lab.

Calculating the mid-range of a set of data.

Determine the mid-range of the following set of data: ΔT / K : 10.33, 10.42, 10.41, 10.60, 10.57, 10.33, 10.49
Aligning the data in increasing order we have:
ΔT / K : 10.33, 10.33, 10.41, 10.42, 10.49, 10.57, 10.60
Mid-range $= \frac{\text{Max value}+\text{Min value}}{2}$
$\Delta T_{\text{Mid-range}} = \frac{10.60 \text{ K}+10.33 \text{ K}}{2} = 10.46 \text{ K}$
Here we have to maintain the number of significant figures in our calculation, so we report the mid-range to 2 decimal places, the same as the data we are using to calculate it. The answer has then been rounded to even.

3.1.3 Exercises calculating the mean and mid-range.

3.1.3.1 Self check on our learning so far.

Determine the mean of the following set of data: $\Delta_r H$ / kJ mol^{-1} : -92.3, -87.4, -90.1

$\bar{\Delta_r H} = -89.9 \text{ kJ mol}^{-1}$
$\bar{\Delta_r H} = \frac{\sum_{i=1}^{N} \Delta_r H_i}{N}$
here $N = 3$, and $\sum_{i=1}^{N} \Delta_r H_i = -269.8 \text{ kJ mol}^{-1}$
$\bar{\Delta_r H} = \frac{-269.8 \text{ kJ mol}^{-1}}{3} = -89.9 \text{ kJ mol}^{-1}$
Note that we have had to round our answer to 1 decimal place, the same as the data we are using to calculate it.

Determine the mean of the following set of data: k / 10^6 s^{-1} : 41.362, 36.52, 37.663, 40.92

$\bar{k} = 39.12 \times 10^6$ s^{-1}
$\bar{k} = \frac{\sum_{i=1}^{N} k_i}{N}$
here $N = 4$, and $\sum_{i=1}^{N} k_i = 156.465 \times 10^6$ s^{-1}
$\bar{k} = \frac{156.465 \times 10^6 \text{ s}^{-1}}{4} = 39.12 \times 10^6$ s^{-1}
Here some of our data has 2 decimal places and some 3 decimal places, our mean needs to reflect the lowest precision in our data, so should have 2 decimal places.

Determine the average of the following data: ΔT / K: 1.205, 1.193, 1.198, 1.204

$\bar{\Delta T} = 1.200$ K
$\bar{\Delta T} = \frac{\sum_{i=1}^{N} \Delta T_i}{N}$
here $N = 4$, and $\sum_{i=1}^{N} \Delta T_i = 4.800$ K
$\bar{\Delta T} = \frac{4.800 \text{ K}}{4} = 1.200$ K
Be careful to record significant zeros (even if your calculator, Excel or Python don't report them).

Determine the mid-range of the following set of data: A: 0.102, 0.097, 0.104, 0.100

$A_{\text{mid-range}} = 0.100$
$A_{\text{mid-range}} = \frac{A_{\text{max}} + A_{\text{min}}}{2}$
Putting the data in increasing order: 0.097, 0.100, 0.102, 0.104
So our $A_{\text{min}} = 0.097$ and our $A_{\text{max}} = 0.104$
$A_{\text{mid-range}} = \frac{0.097 + 0.104}{2} = 0.100$
Note here we have rounded our answer to even, so the 0.1005 rounds down when accomodating the correct number of significant figures. Again remember those significant zeros when writing down your answer.

3.1.3.2 Self assessment questions on mean and mid-range.

3.2 The standard deviation, σ or s.

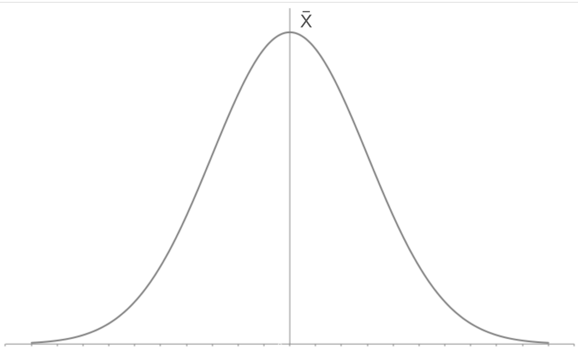
The following two sets of data share the same mean, and the same precision, but they are clearly very different sets of data:

- 9.0, 9.5, 10.0, 10.5, 11.0 $\bar{x} = 10.0$ $x_{min} = 9.0$, $x_{max} = 11.0$
- 9.8, 9.9, 10.0, 10.1, 10.2 $\bar{x} = 10.0$ $x_{min} = 9.8$, $x_{max} = 10.2$

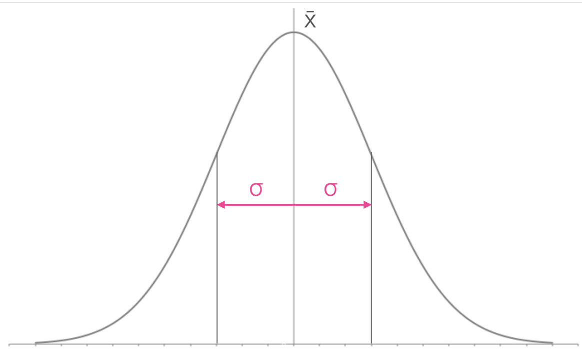
So we need a method to describe the distribution, to help us understand the data.

The standard deviation is represented by the symbol σ or s. These represent two subtly different forms of the standard deviation, the standard deviation of the population (where all possible measurements have been made), *sigma*, and the standard deviation of a sample of the population, s.

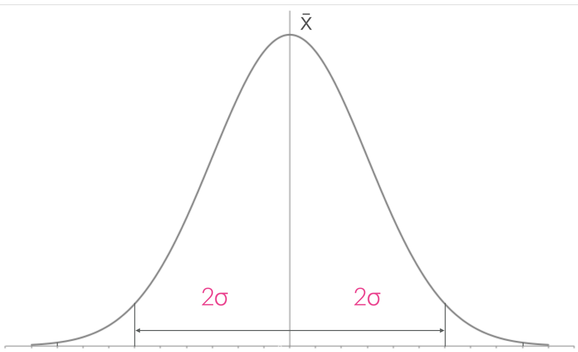
Our Gaussian distribution has a maximum at the mean and a symmetrical distribution around the mean.



(a) The mean is the maximum of the Gaussian distribution.



(b) One standard deviation each side of the mean encompases 68.3 % of the data.



(c) Two standard deviations each side of the mean encompases 95.4 % of the data.

(d) Three standard deviations each side of the mean encompases 99.7 % of the data.

Figure 3.2: The Gaussian distribution and distribution of data described by the standard deviation.

When our data is precise our standard deviation is small.

The standard deviation is a measure of the spread of the data, and therefore a measure of the uncertainty in our measurements. This can be seen in Figure 3.3 which shows multiple Gaussian distributions with different standard deviations. The smaller the standard deviation, the more precise our data is.

The consequence of this larger standard deviation is that the mean value of the data becomes less obvious, and therefore our confidence in the mean value is reduced.

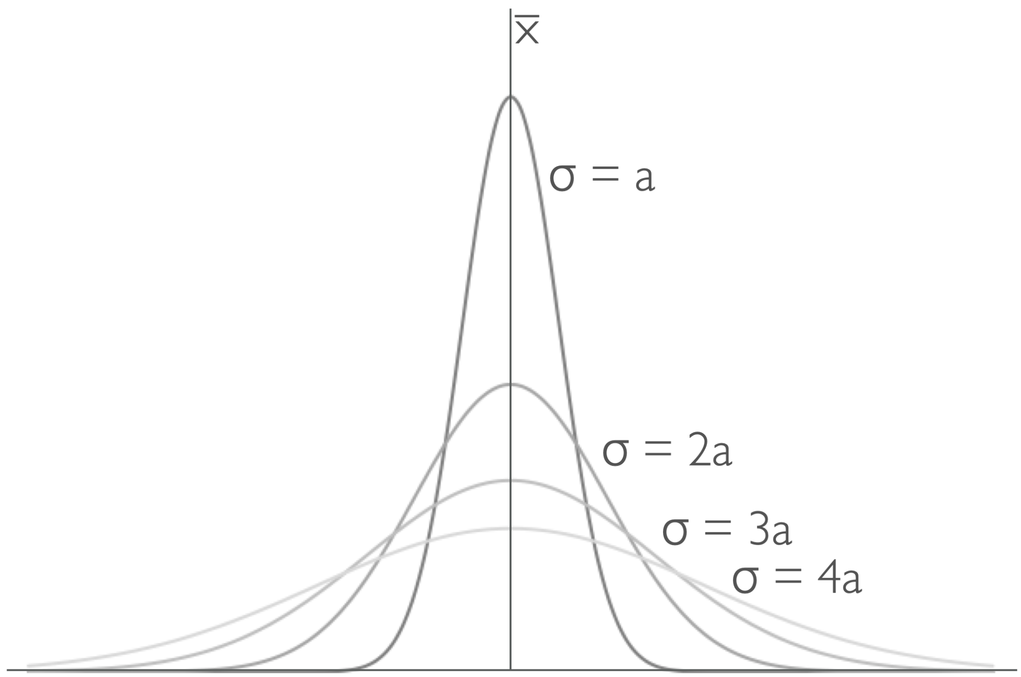


Figure 3.3: The standard deviation is a measure of the spread of the data, the bigger the value the broader the distribution.

If you have learned to calculate the standard deviation previously you will be aware that there are two different methods.

The first method is to calculate the standard deviation of a population, and the second method is to calculate the standard deviation of a sample. The difference between these two methods is that when we are calculating the standard deviation of a sample we divide by $N-1$ instead of N.

We do this because if we have only collected a sample of data, we are not as confident in the mean value of that data as we would be if we had collected all the data. Therefore we need to account for this by dividing by $N-1$ instead of N.

As the number of data points increases the difference between the two methods becomes less significant, and when we have a large number of data points the two methods give very similar results.

In practice we can never measure every possible value of a physical quantity, so we will always be calculating the standard deviation of a sample. Therefore we will always divide by $N-1$

when calculating the standard deviation.

Tip 13: Calculating the standard deviation, s.

In scientific measurements we should always calculate the sample standard devation, s. $s = \sqrt{\frac{\sum_{i=1}^{N}(x_i - \bar{x})^2}{N-1}}$

For completeness the equation for the population standard deviation is:

$\sigma = \sqrt{\frac{\sum_{i=1}^{N}(x_i - \bar{x})^2}{N}}$

although there is no reason we should use this version.

When calculating the standard deviation using Excel you can use the =stdev.s() (=stdev() also works). When using python you can use `statistics.stdev()` or `numpy.std()`.

Calculating the standard deviation of a set of data.

Determine the appropriate standard deviation of the following set of data: ΔT / K : 10.33, 10.42, 10.41, 10.60, 10.57, 10.33, 10.49

$\bar{T} = \frac{\sum_{i=1}^{N} T_i}{N}$

here N = 7, and $\sum_{i=1}^{N} x_i = 73.15$ K

$\bar{\Delta T} = \frac{73.15 \text{ K}}{7} = 10.45$ K

T / K	$\bar{T} - T$ / K	$(\bar{T} - T)^2$ / K^2
10.33	-0.12	0.0144
10.42	-0.03	0.0009
10.41	-0.04	0.0016
10.60	0.15	0.0225
10.57	0.12	0.0144
10.33	-0.12	0.0144
10.49	0.04	0.0016
	$\sum_{i=1}^{N}(x_i - \bar{x})^2 =$	0.0698

3.2.1 Exercises calculating the mean and mid-range.

3.2.1.1 Self check on our learning so far.

4